Software Requirements Specifications

for

Pac-Man Reloaded

Version 1.0

CISC 3140 - Group 2

Brooklyn College

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Splash Screen:

Options:

Presents clear, arcade-style options:
Start Game, Theme, and High Scores.

Layout and Geometry:

Modeled after the classic 1980 arcade title sequence screen, featuring a stark black background and a centered 3:4 vertical layout. The Pac-Man title sits at the top with a 50% gap before the main menu options, and options are spaced evenly at 25% inter-button padding. Interactive options feature blocky retro styling using a dynamic color parameter that adjusts based on the selected display theme.

Start Game Flow:

Selecting Start Game prompts a brief character choice before dropping into the maze:
Select between Mr. Pac-Man or Mrs. Pac-Man (styled after the 1980 arcade sprites, complete with the signature pink bow).

Game Mechanics:

Levels:

There will be three levels, each with increasing difficulty.

Upon the player pressing “Start Game”, level 1 will automatically start, and will continue to the next level until the player either wins or loses. Completing a level advances sequentially to the next level (Level 1 -> Level 2 -> Level 3).

Each level ends once the player has picked up all super and regular pellets throughout the map.

Level completion is worth 1.3x of the total point value in the level. The 1.3x multiplier acts as a scale on the level's total score (Level Subtotal $\times$ 1.3), meaning the base 100

points earned during play gets boosted by a 30% completion bonus (+30 points), yielding a final payout of 130 total points for that stage.

Pellets:

Values for pellets are as follows:

Level 1 = 1 point per pellet
Level 2 = 2 points per pellet
Level 3 = 3 points per pellet

On super pellet pickup:

Ghosts flee from Pac-Man
Inverts colors of ghosts
Adds 50 points to player score. It will give 50 points on all three levels.

Ghosts:

Behavior:

Ghosts spawn within the ghost house in the center of the screen. Blue, Pink, and Orange should be contained in the house, while the red ghost sits above it. When the level starts, the ghosts exit the house and begin a path around the map in random directions.

Scoring:

1st Ghost = 200 pts
2nd Ghost = 400 pts
3rd Ghost = 800 pts
4th Ghost = 1600 pts

Design:

Blinky (red)
Michael Arvanite's head will be in place of the red ghost, Blinky.

Pinky (pink)
Henry T. Berger's head will be in place of the pink ghost, Pinky.

Inky (Blue)
Brain D. Obergfell's head will in place of the blue ghost, Inky

Clyde (Orange)
Barbaralee Diamonstein-Spielvogel's head will be in place of the orange ghost, Clyde.

All headshots for the CUNY trustees can be found at:
www.cuny.edu/about/trustees/

Lives:

When a player hits start game, they start with 3 lives. This means they have 3 attempts to beat all the levels in one single run.

If a ghost catches Pac-Man, the player loses 1 life. Pac-Man and the ghosts immediately respawn back at their default starting positions on the map. The level *does not reset*. Whatever pellets Pac-Man ate before dying stay eaten, and whatever score was earned stays on the board. You just drop back into the starting spot and keep going on that same level.

The life counter does not refill when you complete a level and move to the next one. Example: If you clear Level 1 with 2 lives remaining, you start Level 2 with those same 2 lives, not 3.

The life counter only resets back to 3 under two of the following conditions:
Game Over (0 Lives): You lose your last life, the run ends, and if you press "Retry", you start completely over from Level 1 with a fresh set of 3 lives and 0 score.

Quit to Main Menu: You quit back to the title screen and launch a brand-new game session.

Fruit:

There will be three fruits within the game, each appearing on a separate level.

Cherry - Level 1
Strawberry - Level 2
Orange - Level 3

Spawning:

Fruit will spawn in the center of the map (right below the ghost house) when the player clears exactly 50% of the dots on the map.

Lifespan:

The fruit will last for 10 seconds and disappear if the player doesn’t collect it within the timeframe. It will blink rapidly for the last 3 seconds of its lifespan to alert the player. It vanishes for the rest of that level.

Progression and Points:

Cherry - 100 pts, Strawberry - 350 pts, Orange - 500pts
If the fruit is consumed within the last 3 seconds of its lifespan, an extra 500 points is added to the score.

Controls:

Start (Enter key)
Begins the game when pressed on the start menu. Also acts as a confirm selection during character selection.

Arrow Keys / WASD:
Directional movement (Up, Down, Left, Right / W, A, S, D).

Pause/Freeze (P or Space):
Pressing P or Spacebar halts the main game loop, freezing all movement, timers, and animations in place until pressed again.

Quit/Return to Title (Esc or Q):
Pressing Escape or Q instantly exits the active game run, clears the current session state, and returns the player directly to the title screen.

Quick Restart (R Key):
Pressing R allows players to quickly restart from Level 1 without having to quit all the way back to the main title screen.

Mute Sound Toggle (M Key):
Toggles audio on or off instantly without requiring a settings menu.

Color Theme (T Key):
Toggles through available color palettes/themes on the fly directly during gameplay. Also for navigating themes on main menu

End Game Screens:

High score screen:

Retrieves the top 10 high scores are **l**ocal to that specific browser.

Displays the following:

Rank
Score
Initial

The players in the top 10 will be ranked 1-10, with the player with the highest score ranking first, and the player with the tenth highest score, ranking 10.

“Score” is whatever score the player got.

“Initial” is however the player chose to identify themselves. This should only include three, capital letters. For example: “ABC”, “MAX” , “JJK”, etc.

Clear Level/Win/Lose Screen:

Level Completion Screen (Intermediate Win):
Triggers immediately when all pellets are cleared on Level 1 or Level 2. The overlay displays the current level score, the 1.3 completion bonus, and the updated cumulative total score. It provides navigation options to proceed to the Next Level or return to the Main Menu.

Final Victory Screen ("You Win!"):
Triggers upon clearing all pellets on Level 3 (the final level). The screen displays the final cumulative score and checks if the score qualifies for the local Top 10 leaderboard (prompting for 3-character initials if qualified). It provides options to Retry / Play Again or return to the Main Menu.

Game Over Screen:
Triggers when the player's life total reaches 0. The overlay displays the final score and the highest level reached during the run, checks for local high-score qualification, and offers options to Retry (starting a new session from Level 1 with 3 lives and 0 score) or return to the Main Menu.

Visual and Sound:

Fonts:

All the text in the game should be “Press Start 2P” from Google fonts. (See below) Otherwise, it should default to a monospace font. It should be all white, except for the “Ready!” text that appears at the start of every level, which should be bright yellow.

fonts.google.com/specimen/Press+Start+2P

Theme:

Color selection (Changes background color)
1 colorblind mode (Changes text for color)

Audio:

Pacman's mouth motion must be in tandem with his iconic “wa” sound
A Pac-Man song is to be played throughout the level as players complete it
A sound on super pellet pickup and a change of music to indicate to the player that they can begin chasing the “ghost”